

BIRDS

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IN THE LIVING WORLD, only birds, insects, and bats are capable of powered flight. Birds are the largest and fastest of these flying animals, and are the only ones that have feathers. There are about 9,000 species of bird, and they live in a huge range of different habitats – from deserts to the open oceans. They eat a variety of food, which they find mainly by sight. All birds reproduce by laying eggs. Most look after their young until they can fend for themselves.



Pigeon in flight

Flight feathers are spread out as the bird prepares to land.

Wings almost touch during the upstroke.

Fanned tail feathers act as a brake.

Feet are held against the body during flight.

Bird features

Birds have a lightweight skeleton and their feathers give them a smooth outline, which helps them move easily through the air. They do not have any teeth, but they have a hard beak instead. Birds use their beaks for eating, and also for many tasks that other animals carry out with their front legs and feet, such as grasping items, or tearing up food.

Legs and feet

A bird's feet and lower legs are usually covered with scales. Muscles that move them are close to the body. The feet are shaped according to their use.

Beak

A bird's beak is covered with keratin – the same substance that makes up human fingernails. The keratin keeps growing so that the edges of the beak do not wear away.

Wings

The bones in a bird's wing are similar to those in a human arm. Most birds use their wings to fly. Strong muscles pull the wings downward when the bird flies; other muscles fold them up when not in use.



Internal air space with reinforcing struts

Bone structure

Most of the larger bones of a bird are hollow, which saves weight. They contain air spaces that connect to the special air sacs the bird uses when it breathes. Some diving birds have solid bones to make diving easier.

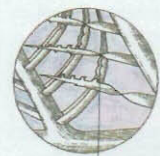
Skeleton

Birds have fewer bones than reptiles or mammals, and many of the bones are fused together. A large flap called the keel sticks out of the breastbone and anchors the muscles that power the wings.



Feathers

Birds use their feathers to fly, and also to keep warm and dry. Each feather is made of fine strands called barbs that carry rows of smaller barbules. In some feathers, the barbules lock together with hooks to produce a smooth surface needed for flying through the air. In others, they stay partly or fully separate. These feathers are soft and fluffy for warmth.



Microscopic hooks lock barbules together.

A hollow quill anchors the feather in the bird's skin.

Macaw flight feather

Curved tip with interlocking barbules

Continuous curved surface

Central quill



Breeding colours

Male birds often have bright colours which attract mates. In some species, these colours disappear at the end of the breeding season when the birds moult and a new set of feathers grows. In other species, such as pheasants, the colours are permanent.

Down feathers

These short fluffy feathers do not have hooked barbs. They form an insulating layer next to a bird's skin. They trap air, which helps to stop heat from escaping from the bird's body.

Body feathers

The tips of these feathers overlap like tiles on a roof, giving the bird a smooth shape. The fluffy base of each feather is close to the body and helps to save heat.

Flight feathers

These feathers are strong but flexible. They provide lift when the bird is airborne. Birds have to preen them carefully to keep them in good condition.

Tail feathers

A bird uses its tail feathers for steering and braking. Some male birds have long or brightly coloured tail feathers. These play an important part in courtship.